Low-Q point-addition circuit `b6f2b0a`

Source-grounded anatomy of submission `b6f2b0a` / commit `e6d9fa9`

Submitted operating point

$Q = 825$ logical qubits
$T = 489{,}161{,}900$ Toffoli gates
$Q \times T = 403{,}558{,}567{,}500$

Inversion architecture

Register-shared, shrunken Proos–Zalka-style EEA
1,479 scheduled reversible steps
No dialog transcript and no dialog-GCD replay

Q825-specific change

Remove one persistent ℓ_q lane
Reconstruct its high bit implicitly
Use an eight-scratch split-five kernel

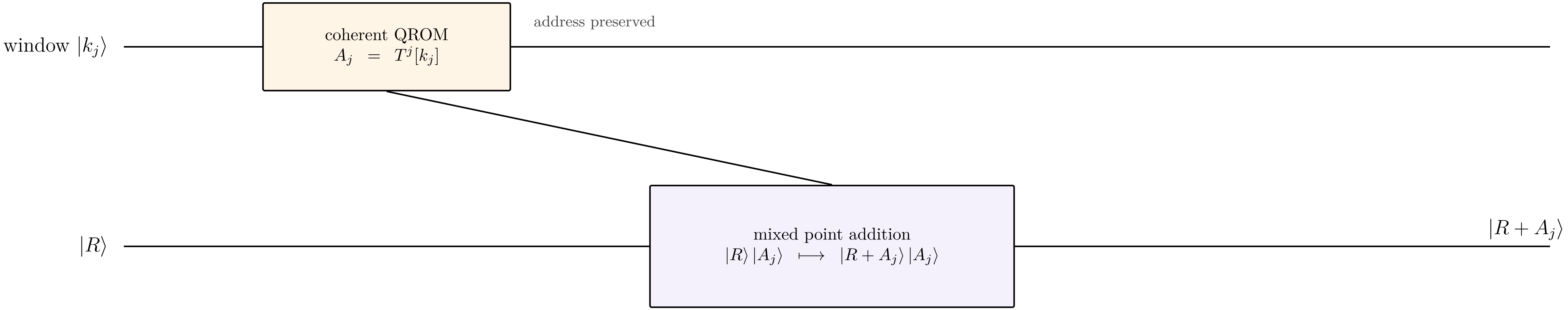
What this circuit optimizes. The benchmark circuit maps a quantum accumulator $R = (x_R, y_R)$ and a classically specified addend $A = (x_A, y_A)$ to $R' = R + A$ over `secp256k1`. The low-Q route spends substantial nonlinear work to minimize the maximum number of simultaneously live logical qubits.

Evidential status. The public row is marked *rejected*, but its note reports an isolated replay of all 9,024 challenge tests with zero classical, phase, and ancilla failures. The seven-bit quotient-length promise is tied to that finite validation support; the deck does not promote it to an all-input proof or a full windowed-Shor resource estimate.

The Q825 circuit is analyzed in an isolated restoration and is separate from the `8e9c9a2` product-score deck.

Where the low-Q kernel sits in windowed Shor

The submitted source implements one mixed addition; coherent point selection is outside its interface.



Challenge interface

x_A and y_A arrive as classical bits. The circuit may materialize them transiently, use classical-control specialization, and release them before the inversion peak.

Full windowed interface

A_j is selected coherently by k_j. Its coordinates are quantum data entangled with the address until QROM unlookup, so classical-only savings do not transfer automatically.

Interpretation

The deck explains the submitted point-addition artifact. A complete Shor integration requires a separately validated lookup/use/unlookup construction and compatible accounting.

Affine mixed addition implemented by `b6f2b0a`

$R = (x_R, y_R)$, $A = (x_A, y_A)$, and $R' = R + A$ in the generic case

In-place register updates

Input: $X = x_R$, $Y = y_R$; classical addend $A = (x_A, y_A)$; $x_R \neq x_A$.

1. $Y \leftarrow y_A - Y$ — $\triangleright$ $Y = d_y = y_A - y_R$
2. $X \leftarrow x_A - X$ — $\triangleright$ $X = d_x = x_A - x_R$
3. $\lambda \leftarrow Y X^{-1}$ — $\triangleright$ $\lambda = d_y / d_x$
4. $Y \leftarrow 0$ — $\triangleright$ reuse Y as scratch
5. $X \leftarrow \lambda^2 - x_R - x_A$ — $\triangleright$ $X = x_{R'}$
6. $X \leftarrow x_A - X$ — $\triangleright$ $X = d'_x = x_A - x_{R'}$
7. $Y \leftarrow \lambda X$ — $\triangleright$ $Y = d'_y = \lambda(x_A - x_{R'})$
8. $\lambda \leftarrow 0$ — $\triangleright$ $\lambda = d'_y / d'_x$ witnesses cleanup
9. $X \leftarrow x_A - X; \quad Y \leftarrow Y - y_A$ — $\triangleright$ $X = x_{R'}$, $Y = y_{R'}$

Slope convention

The source uses

$$d_x = x_A - x_R, \qquad d_y = y_A - y_R, \qquad \lambda = d_y d_x^{-1}.$$

Reversing both differences leaves the usual slope unchanged.

Coordinate identities

$$x_{R'} = \lambda^2 - x_R - x_A, \qquad y_{R'} = \lambda(x_A - x_{R'}) - y_A.$$

The implementation overwrites X and Y rather than allocating separate output coordinates.

Two division lifecycles

The first EEA traversal produces λ from (d_x, d_y). After the coordinates are formed, the identity $\lambda = d'_y / d'_x$ permits a second EEA traversal to erase λ without retaining the first inversion history.

Point-addition pipeline and source correspondence

Both 825-qubit peaks occur inside the register-shared EEA stages.

$X : |x_R\rangle$

$Y : |y_R\rangle$

1. Differences

$X = x_A - x_R = d_x$
$Y = y_A - y_R = d_y$

classical addend paths

2. Forward divide

$\lambda = d_y/d_x$

restore d_x, d_y

peak: 825 qubits

3. Zero and square

erase d_y

$X = \lambda^2 - x_R - x_A$

Y becomes scratch

4. New differences

$X = x_A - x_{R'}$
$Y = \lambda X$

alternate witness

5. Cancel divide

recompute witness

erase λ

peak: 825 qubits

6. Restore output

$X = x_{R'}$
$Y = y_{R'}$

remove pad lanes

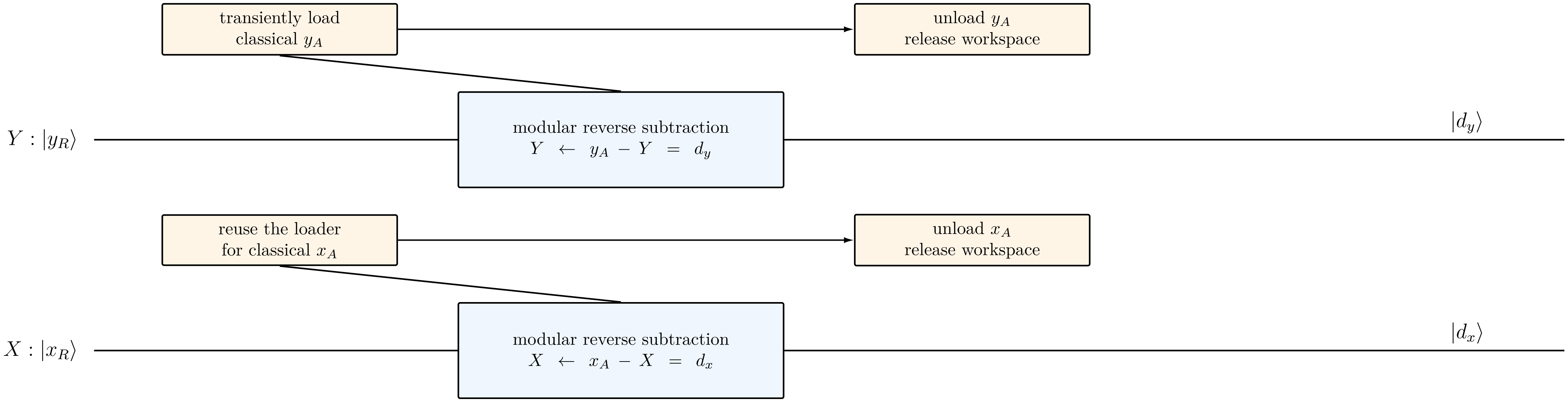
Source sections and live-width maxima

Differences	Forward divide	Zero and square	New differences	Cancel divide	Restore output
`ec3.dy_build`, `ec3.dx_build`	`ec3.inv_fwd`	`ec3.dy_zero`, `ec3.new_x`	`ec3.alt.new_dx`, `new_dy`	`ec3.alt.cancel`	`ec3.alt.new_x_restore`, `new_y_restore`
max $Q = 788$	max $Q = 825$	max $Q = 789$	max $Q = 788$	max $Q = 825$	max $Q = 789$

Count-only source instrumentation reproduces the submitted Q and structural T exactly.

Stage 1: build d_y and d_x with low-pressure classical loads

$d_y = y_A - y_R \pmod p$, followed by $d_x = x_A - x_R \pmod p$



Serialized materialization

The two 256-bit classical coordinates are not simultaneously materialized. A transient quantum loader is used for one subtraction, restored to zero, and then reused.

Pseudo-Mersenne arithmetic

With $p = 2^{256} - (2^{32} + 977)$, overflow and borrow corrections are folded through the small constant $2^{32} + 977$ instead of a generic full-width modular reduction.

Scope

These loads exploit the challenge's classical-addend interface. In a coherent QROM-based Shor call, x_A and y_A are quantum data and need a different implementation.

Stage 2: register-shared shrunken-PZ division

A reversible extended Euclidean state machine computes $\lambda = d_y d_x^{-1}$ without a dialog transcript.

initialize
$r = p, r' = |d_x|$
$t = 1, t' = 0$

$\rightarrow$

1,479 scheduled EEA steps
update packed remainders, coefficients,
quotient, lengths, and control bits

$\rightarrow$

terminal inverse
rotate and sign-correct t'
to obtain d_x^{-1}

$\rightarrow$

canonical multiply
$\lambda \leftarrow d_y\, d_x^{-1} \pmod p$

$\rightarrow$

reverse all EEA steps
restore d_x
clear work state

No recorded dialog

Dialog-GCD stores a compressed branch transcript and later replays it. Here the circuit carries the evolving Euclidean state itself in two packed work words and reverses the same scheduled gate sequence. The cost is high T; the benefit is very low resident width.

Register-shared core

$$W_1 = (t \mid q \mid r), \qquad W_2 = \mathrm{rot}_{\ell_s}(t' \mid r').$$

Length metadata $\ell_t, \ell_q, \ell_{t'}, \ell_{r'}, \ell_s$ identifies the active subranges; phase, sign, and iteration bits control the reversible microsteps.

Passenger control

Only one full coordinate accompanies the EEA peak. The preserved clean top lane of the passenger is repeatedly borrowed as temporary storage and restored before the terminal multiplication or reverse schedule needs it.

Source: `register_shared_divide_forward`, `register_shared_forward`, *and* `register_shared_reverse`.

Stage 2 detail: the packed Euclidean state machine

The circuit incrementally forms quotient bits and updates a remainder/coefficient pair.

Packed invariants

$$W_1 = t \parallel q \parallel r, \qquad W_2 = \mathrm{rot}_{\ell_s}(t' \parallel r').$$

For $n = 256$, each physical work word has $n + 3 = 259$ lanes, while

$$\ell_t + 1 + \ell_q + \mathrm{bitlen}(r) \le n + 3, \qquad \mathrm{bitlen}(t') + \ell_{r'} \le n + 3.$$

One scheduled microstep

A shift counter ℓ_s selects an aligned slice of r'. The reversible core conditionally forms

$$r \leftarrow r \mp 2^{\ell_s} r', \qquad t' \leftarrow t' \mp 2^{\ell_s} t,$$

toggles the corresponding quotient/sign information, and updates phase bits. When the quotient and shift counters return to zero, (r, t) and (r', t') are exchanged.

Fixed-length schedule

The state machine always executes 1,479 steps. After the Euclidean computation reaches $r' = 0$, reversible padding steps preserve a fixed circuit shape. At the terminal boundary,

$$r = 1, \quad r' = 0, \quad q = 0,$$

and the signed coefficient t' represents the modular inverse.

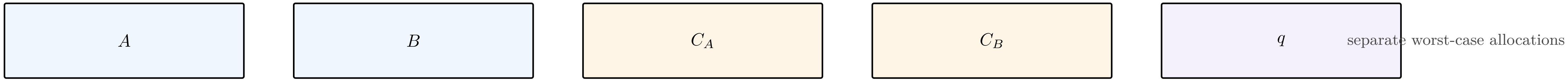
Why this differs from dialog-GCD. No compact branch word is stored for later interpretation. Instead, the active remainders, coefficients, quotient fragment, and their length metadata remain the reversible state. The inverse circuit is obtained by traversing the scheduled microsteps in reverse order.

The production route combines exact reversible microkernels with support-derived width schedules and metadata promises. Consequently, the architectural identities above are broader than the finite-support evidence for every selected truncation and hosting decision.

Stage 2 detail: register sharing and dynamic width

The principal low-Q gain comes from assigning physical lanes by lifetime rather than by logical register name.

Naive logical layout



Production packed layout

Sliding ownership boundaries

As a remainder loses high bits, the paired coefficient acquires low-order space. Length registers move the logical boundary without keeping two independent 256-bit allocations.

Support-tuned schedule

Per-step active ranges, comparison widths, and quotient caps are selected from a thin schedule. High lanes known to be inactive are released or loaned to local predicates and carry chains.

Reversible lending discipline

A borrowed lane may host temporary information only while its logical owner is inactive. Each microkernel must restore the lane before the owner or a neighboring peak stage becomes live again.

The Q826 → Q825 change: omit one persistent ℓ_q bit

This submission saves one peak lane by reconstructing the high bit only where it is consumed.

Before: eight stored quotient-length lanes

$$\ell_q = (\ell_{q,7}\,\ell_{q,6}\cdots\ell_{q,0})_2.$$

The Q826 route keeps all eight bits resident while three disjoint host windows reuse other lanes. The persistent $\ell_{q,7}$ lane participates in the peak even though the submitted support never observes $\ell_q \geq 128$.

After: seven stored lanes plus implicit reconstruction

$$\ell_q = (\ell_{q,6}\cdots\ell_{q,0})_2, \qquad \ell_{q,7} \text{ is not persistently allocated.}$$

When a split-five length kernel needs the omitted bit, it evaluates an equivalent zero-prefix predicate directly and writes only the final controlled effect.

Implicit high-bit identity

For a 259-bit source, let Z_k mean that every source bit in positions $k-1$ and above is zero. The 2^7 bit of the source bit length is

$$h_7 = Z_{256} \oplus Z_{128}.$$

Thus the high bit can control a target without being materialized.

Eight-scratch split-five kernel

The prior serial split-five route used nine scratch lanes. Q825 materializes only length bits $2^4, 2^5, 2^6$ and reconstructs 2^7 inside a one-short subtraction/MCX path, reducing the qualified scratch requirement from nine lanes to eight.

Boolean factorization

Terms sharing h_7 are factored, for example

$$h_7 y \oplus h_7 b = h_7(y \oplus b),$$

and reachable zero-prefix relations simplify the final-borrow controls. This replaces a stored bit with recomputation and additional multi-controlled operations.

The committed support audit reports $\max \ell_q = 22$ over 18,048 inversion factors. This supports the seven-bit route on the benchmark corpus; it is not, by itself, a universal proof that every field input satisfies $\ell_q < 128$.

Inherited low-Q optimization stack

Q825 composes many earlier width reductions; the omitted ℓ_q lane is only the final step.

Three Q826 host windows

Separate substages lend inactive t' or ℓ_s lanes to remainder, coefficient, and rotated-swap microkernels. Their lifetimes are disjoint, so all three one-qubit savings can coexist.

Metadata relocation

Direct swap metadata, a relocated coefficient counter, a fixed parity bit for ℓ_s, and reuse of ℓ_q as old-remainder metadata reduce the persistent control-register footprint.

Predicate and comparison narrowing

Prefix bit-length circuits operate only on active ranges. Truncated guards, direct selectors, and borrowed underflow lanes avoid full-width comparators when the route supplies a tighter bound.

Paired source complements

Adjacent bit-length kernels retain a complemented source across a source-disjoint use block, allowing the middle pair of X layers to commute and cancel instead of being emitted twice.

Carry and scratch reuse

Zero carries, comparator chains, raw-prefix lanes, the preserved passenger top bit, and clean coefficient scratch are reused across nonoverlapping local kernels and restored at each ownership boundary.

Measurement-assisted cleanup

Temporary logical-AND and carry ancillas are measured in the X basis and corrected by classical phase feedback, avoiding selected reverse-Toffoli cleanup while releasing qubits promptly.

Design principle: the route minimizes the union of simultaneously live information, not the sum of nominal register widths. Most individual changes matter only when they relieve a co-binding peak.

Producing λ while preserving d_x and d_y

Terminal constants are released before multiplication, and the passenger is reconstructed after reverse EEA.

forward EEA
terminal packed state

→

release known terminal values
retain only inverse coefficient

→

$\lambda \leftarrow d_y d_x^{-1}$
canonical modular multiply

→

ghost and free d_y
rebuild terminal constants

→

reverse EEA
restore d_x

Terminal teardown

At convergence, most EEA fields are constants. The implementation toggles them to zero and releases their lanes before allocating the 257-lane λ result. This prevents the complete terminal EEA pack from overlapping the modular multiplication.

Passenger ghosting

After λ is formed, each d_y lane is Hadamard-measured and represented by a classical ghost record. The quantum lanes are freed while the EEA runs backward. The source later recomputes

$$d_y = \lambda d_x$$

and resolves the stored phase corrections against the reconstructed register.

Top-lane hosting

Canonical field values carry a clean 257th lane. During selected reverse-schedule substages, the clean λ or d_y top lane hosts omitted metadata or predicate scratch, then returns to zero before the modular multiply or API boundary.

Coordinate update and alternate-witness cleanup

The same X and Y registers are recycled from input coordinates to output coordinates.

erase d_y
$Y \leftarrow d_y - \lambda d_x = 0$

→

form new x coordinate
$X \leftarrow \lambda^2 - x_R - x_A = x_{R'}$

→

reuse differences
$X \leftarrow x_A - x_{R'} = d'_x$
$Y \leftarrow \lambda d'_x = d'_y$

→

cancel slope ancilla
recompute $d'_y/d'_x = \lambda$
resolve and erase λ

→

restore output
$X = x_{R'}$
$Y = y_{R'}$

Horner multiply-and-accumulate
Products such as λ^2 and $\lambda d'_x$ are accumulated with a bit-serial Horner loop. Pseudo-Mersenne folds keep each intermediate near field width; a retained one-qubit reduction flag is consumed by the matching inverse or by a canonical-output cleanup.

Zero-d_y/new-d_x route
Once $d_y = \lambda d_x$ has served as the slope numerator, it is erased. Its 257 lanes become local loading and multiplication scratch while X moves through x_R, $x_{R'}$, and d'_x. This avoids a separate full-width difference register.

Second EEA lifecycle
The cancel phase ghosts λ, computes the inverse of d'_x, recreates the quotient $d'_y(d'_x)^{-1}$, resolves the ghost, undoes the temporary multiply, and reverses the EEA. This is not a replay of the first GCD path; it is a fresh reversible division on the alternate witness.

Measured resource profile of the restored source

Count-only instrumentation reproduces the submission and explains the low-Q/high-T tradeoff.

Whole circuit

$Q = 825$

$T = 489{,}161{,}900$

$QT = 403{,}558{,}567{,}500$

Forward divide

peak $Q = 825$

$T = 225{,}376{,}748$

46.07% of total T

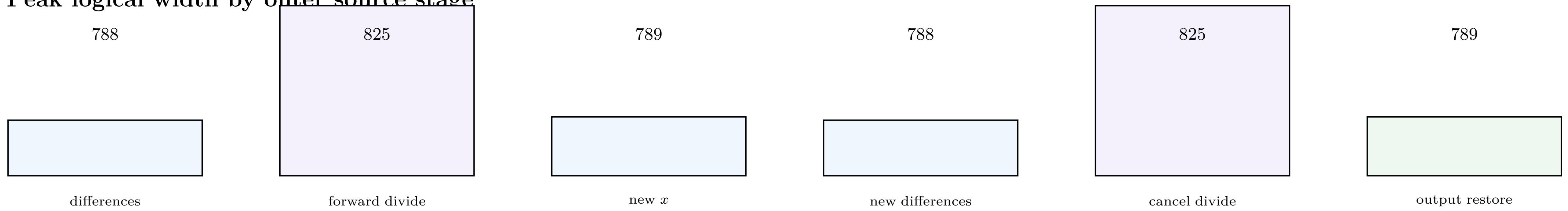
Cancel divide

peak $Q = 825$

$T = 225{,}376{,}748$

46.07% of total T

Peak logical width by outer source stage



Inversion-dominated cost. The two EEA stages contribute 450,753,496 Toffolis, or 92.15% of the circuit total. The route is therefore a hard-width witness rather than a product-score-efficient operating point.

Marginal Q825 trade. Relative to the cited Q826 parent point $(826, 479{,}337{,}428)$, Q825 saves one qubit while adding 9,824,472 Toffolis (2.05%), increasing $Q \times T$ by approximately 1.93%.

Validation record and evidential boundaries

The source contains substantial component and composition checks, but the strongest claims remain finite-support.

Reported evaluator replay

9,024 Fiat–Shamir test cases
zero classical-output failures
zero phase failures
zero ancilla-garbage failures
maximum observed $\ell_q = 22$

Q825 component checks

Ten-control one-short MCX exhaustively checked over 262,144 basis states. The bounded split-five component check covers source lengths $0 \ldots 259$, representative source classes, boundary forms, control values, phase, inverse, and scratch restoration.

Composition evidence

The submission note reports Q826/Q825 comparison at 425,952 internal boundaries, 425,984 semantic snapshots, and 64 bit-sliced measurement branches per factor, together with forward/inverse cleanup checks.

What is established

A concrete gate stream with the reported Q and structural T; clean behavior on the pinned validation corpus; source-level mechanisms for the omitted lane; and bounded verification of the relevant microkernels and selected compositions.

What remains separate

Universal correctness of every support-tuned width promise; completeness of the public archive; coherent QROM compatibility; full windowed-Shor cost; physical error correction; and a proof that the 825-qubit point is globally optimal.

The public status "rejected" should not be read as a correctness diagnosis. In the challenge workflow, a clean but non-promoted hard-Q point may remain rejected because it does not improve the primary leaderboard score.

Source map and reproduction notes

The analysis is tied to submission `b6f2b0a` and source commit `e6d9fa9`.

Point-addition driver

`src/point_add/trailmix_port/ec/point_add.rs`
`ec_add_inplace_shrunken_pz`

Builds d_x, d_y, invokes the register-shared divider, recycles X, Y through the coordinate formulas, invokes alternate-witness cancellation, and restores the 256-bit interface.

Register-shared EEA

`src/point_add/trailmix_port/inversion/`
`register_shared_eea.rs`
`register_shared_eea_reference.rs`

Defines the packed state, scheduled forward/reverse microsteps, host windows, Q825 implicit-bit kernel, and component/composition checks.

Route configuration and instrumentation

`src/point_add/trailmix_port/mod.rs`

Source-bakes the low-Q flags and tail nonce, allocates the benchmark registers, reports support statistics, and provides count-only stage/peak instrumentation.

Arithmetic substrate

`trailmix_port/rfold_mbu.rs` and `trailmix_port/arith/`

Implements canonical and rfold modular multiplication, pseudo-Mersenne correction, comparison, prefix, multi-control, carry, and measurement-assisted cleanup primitives.

Restore with `ecdsafail reset b6f2b0a`. For a non-emitting structural recount, run the release builder with `POINT_ADD_COUNT_ONLY=1`, `TRACE_PHASE_OPS=1`, `TRACE_PHASE_ACTIVE=1`, and `TRACE_STRUCTURAL_COUNT=1`.

Primary references for the architecture: Proos–Zalka reversible ECDLP arithmetic and the later register-sharing formulation cited by the source; the production implementation is the audited challenge artifact above.